

NORILSK AP, Russian Federation

WMO: 319811

Lat: 69.311N

Lon: 87.332E

Elev: 175

StdP: 99.24

Time Zone: 7.00 (E07)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.2	-38.1	-44.9	0.0	-40.5	-42.1	0.1	-38.0	19.9	-19.2	17.7	-18.3	4.4	50	1.027

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	25.1	15.8	22.9	15.2	20.1	14.0	17.2	22.3	16.0	21.0	14.7	19.5	5.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.0	18.9	13.9	10.1	17.7	12.2	9.0	16.3	49.3	21.5	45.5	21.2	41.4	19.6	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	15.0	13.4	DB	-43.4	27.6	3.5	2.6	-45.9	29.5	-48.0	31.0	-50.0	32.5	-52.6	34.4	(4)
(5)			WB	-43.4	19.0	3.5	1.8	-46.0	20.3	-48.0	21.4	-50.0	22.4	-52.5	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.5	-22.6	-24.5	-19.2	-11.3	-4.2	8.4	14.2	9.8	4.6	-6.7	-17.6	-21.6
	DBStd	14.90	7.03	7.94	8.75	8.05	5.55	5.15	4.67	4.02	4.35	6.29	6.97	8.00
	HDD10.0	6614	1009	966	906	639	442	92	12	52	170	518	828	980
	HDD18.3	9437	1268	1199	1164	889	698	302	146	265	413	777	1078	1238
	CDD10.0	243	0	0	0	0	2	43	143	47	8	0	0	0
	CDD18.3	23	0	0	0	0	0	3	19	1	0	0	0	0
	CDH23.3	159	0	0	0	0	0	20	133	6	0	0	0	0
	CDH26.7	21	0	0	0	0	0	2	19	0	0	0	0	0
Wind	WSAvg	6.4	7.8	6.6	6.7	6.7	6.2	5.5	5.6	5.4	5.8	6.3	6.3	7.5
Precipitation	PrecAvg	871	72	64	62	62	61	59	61	80	69	110	85	86
	PrecMax	1110	145	120	131	121	105	105	160	134	143	178	163	143
	PrecMin	431	13	13	11	3	12	8	10	29	30	33	18	8
	PrecStd	139	34	28	30	27	22	25	33	24	25	37	36	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.9	-5.2	-0.9	5.0	19.0	26.0	28.9	24.1	19.2	9.8	0.2	-4.8
		MCWB	-4.9	-5.4	-1.2	2.0	11.7	15.6	17.1	16.2	13.8	8.3	0.2	-4.8
	2%	DB	-9.1	-7.9	-3.8	3.2	7.2	22.9	26.9	21.0	15.0	4.1	-2.0	-6.2
		MCWB	-9.2	-8.0	-4.1	1.5	4.6	14.2	16.4	15.0	10.8	3.7	-2.0	-6.4
	5%	DB	-11.1	-10.9	-6.1	2.0	4.2	19.2	24.9	18.1	12.8	2.8	-5.1	-8.2
		MCWB	-11.3	-11.0	-6.5	0.8	2.3	12.0	15.9	13.2	9.6	2.4	-5.6	-8.6
	10%	DB	-13.9	-14.0	-8.0	0.1	2.9	16.8	22.1	16.0	10.8	1.0	-7.9	-10.9
		MCWB	-14.1	-14.3	-8.7	-0.5	1.4	11.0	15.5	12.7	8.8	0.7	-8.4	-11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.1	-5.6	-1.3	2.6	11.5	17.1	19.8	17.1	14.5	8.5	0.1	-4.9
		MCDB	-5.0	-5.5	-0.9	3.9	18.9	23.6	25.1	22.1	18.8	9.4	0.1	-4.9
	2%	WB	-9.6	-8.1	-4.0	1.7	4.6	14.9	18.0	15.6	12.0	3.9	-2.1	-6.7
		MCDB	-9.5	-7.9	-3.8	3.0	6.7	21.0	22.9	19.6	14.0	4.1	-2.0	-6.6
	5%	WB	-11.5	-11.0	-6.6	0.7	2.6	13.0	17.0	14.2	10.1	2.2	-5.7	-8.7
		MCDB	-11.3	-10.9	-6.3	1.7	3.8	18.4	22.0	17.2	11.7	2.4	-5.2	-8.4
	10%	WB	-14.3	-14.3	-8.6	-0.5	1.4	11.2	15.9	13.0	8.8	0.4	-8.4	-11.1
		MCDB	-14.1	-14.1	-8.2	0.3	2.4	15.9	21.0	15.5	10.6	0.7	-7.9	-10.9
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.4	7.2	7.1	5.7	7.4	7.9	6.5	5.2	4.6	5.9	6.0
		MCDBR	7.5	8.4	6.8	5.5	6.4	11.6	10.3	9.3	7.5	3.8	7.3	7.9
		MCWBR	7.5	8.4	6.8	4.6	4.8	6.0	4.8	4.9	4.7	3.4	7.4	7.3
	5% WB	MCDBR	7.5	8.5	6.9	5.1	6.0	10.6	9.0	8.2	6.5	3.7	7.6	7.5
		MCWBR	7.5	8.5	7.0	4.5	4.7	6.7	5.0	4.9	4.7	3.5	7.7	7.5
Clear-Sky Solar Irradiance	taub	0.212	0.247	0.303	0.328	0.359	0.374	0.402	0.394	0.326	0.249	0.199	N/A	
	taud	1.847	1.953	1.988	2.023	2.054	2.305	2.262	2.233	2.356	2.063	1.874	N/A	
	Ebn at Noon	18	467	677	791	810	818	772	723	687	463	16	0	
	Edh at Noon	5	48	91	120	133	108	109	100	66	40	4	0	
All-Sky Solar Radiation	RadAvg	0.01	0.36	1.92	4.08	5.69	5.35	5.01	3.13	1.59	0.50	0.06	0.00	
	RadStd	0.00	0.05	0.21	0.33	0.47	0.47	0.71	0.36	0.18	0.09	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page